Tianhao Li

(+86) 13121515269 litianhao.life tianhao.li2@duke.edu LinkedIn Google Scholar Orcid Github

Experience

Beijing TopSec Network Security Technology **March 2024 – July 2024 (5 mos)**

Security Researcher @ Department of Innovative Technology Research *Beijing, China*

- **Duties**: **(i)** Supported automated red teaming research on LLMs, exploring techniques like model fine-tuning and RLHF. Researched innovative technologies for prompt obfuscation, variation, and failure detection. Improved automation capabilities and vulnerability discovery efficiency in red team testing, reducing reliance on security experts; **(ii)** Assisted in implementing a prototype system for machine learning model security assessment projects. Worked on debugging adversarial and poisoning attacks, and developing the backend evaluation module of the system.

NSFOCUS Technologies Group Co.,Ltd **September 2023 – March 2024 (7 mos)**

Security Research Intern @ Innovation Institute Dubhe Lab *Beijing, China*

- **Duties**: **(i)** Design core algorithms and develop a prototype for NSFOCUS LLM Security Assessment System (LSAS); **(ii)** Enhance the security & privacy performance of NSFGPT and other private LLMs through offensive testing, i.e. red teaming; **(iii)** Conduct literature reviews encompassing research papers and Gartner's consulting reports on trustworthy AI, enriching the threat intelligence database; **(iv)** Write white paper on secure LLM.
- **Impacts**: The release of LSAS, SecLLM NSFGPT, and white paper Enhancing Network Security with SecLLM.
- **Mentors**: Dr. Xingkai Wang, Dr. Wenmao Liu

High-Tech Innovation Institute **July 2022 – July 2023 (1 yr 1 mo)**

Research Intern @ VR & AR Technology Innovation Center *Beijing, China*

- **Duties**: **(i)** Participate in the national key research and development program; **(ii)** Research on gait-based intelligence, surveillance and reconnaissance (ISR) system; **(iii)** Research on classification of human emotion based on electroencephalogram (EEG) and Swin-Transformer; **(iv)** Research on early warning mechanism for aviation safety based on QAR stream data; **(v)** Research on threat model and mitigation methods in AI-XR-based Metaverse.
- **Impacts**: Co-authored 7 papers indexed in SCI/EI, secured victory in 4 national professional contests, delivered 3 oral presentations and presented 1 poster at international conferences.
- **Mentors**: Prof. Dr. Lijun Wang, Prof. Dr. Zhengping Li, Assoc. Prof. Dr. Ying Li

Chenchuang Technology Development Co.,Ltd **January 2022 – March 2022 (3 mos)**

C++ Software Development Intern @ Technical Development Department *Chengdu, China*

- **Duties**: **(i)** Develop a real-time streaming video decoding and object detection module using WASM, OpenCV, YOLOv5; **(ii)** Design and implement a secure key exchange mechanism based on SM2 and SM3, encrypted RTSP video streams with SM4, and employed SM4 for encrypting stored logs of cloud-based object detection data, ensuring secure data transmission and storage.
- **Impacts**: Contributed to the research, development and release of airfield runway foreign object debris (FOD) detection and management system.

Projects (Selected)

LLM Security Vulnerability Assessment System **September 2023 – June 2024 (Expected)**

Algorithm Researcher, Full-stack Developer Opening Defense | Mid Inspection | Thesis Defense

- **As algorithm researcher**: Designed probes and detectors for diverse vulnerabilities (e.g., adversarial attack, jailbreak, DAN). Implemented multi-modal LLM adoption for testing across text, image, and audio domains. (LLM red teaming tool)
- **As full-stack developer**: Developed Python and Django-based B/S platform supporting ModelOps, model monitoring, and privacy protection features. (Practice of Gartner's AI TRiSM conceptual framework)

Gait-based Intelligence, Surveillance, Reconnaissance (ISR) System **May 2023 – August 2023**

Team Leader, Algorithm Engineer, Backend Developer Demo Video 1 | Demo Video 2 | About Page

- **As team leader**: Assigned team roles based on strengths, led system security capacity building, led technical architecture design, and prepared technical report.
- **As algorithm engineer**: Implemented gait recognition pipeline, trained and evaluated models using PyTorch, and developed system security features including authentication, tracking, and early warning mechanisms for physical attacks.
- **As backend developer**: Developed encryption and network modules utilizing SM2/SM3/SM4, including secure key distribution and data encryption/decryption on public channels.

Open Source Activities

- leondz/garak(Generative AI Red-teaming Tool Kit, Metasploit in the field of LLM Security. My main contributions include multi-modal capabilities and the development of intelligent red teaming capabilities.):
  - **Merged**: Python interpreter version bug fix#296; Replicate LLM generator exception error fix#401; Add exception handling logic to the Detector base class#566; Multi-modal Jailbreaking Attack on LLaVA #587 (Make it the **world's first** tool with multimodal red teaming capabilities.). (v0.9.0.9, v0.9.0.11, v0.9.0.13)
  - **Ongoing**: Prompt Architecture Enhancement for Better Multi-modal Red Teaming (#658); Network Proxy Feature For Generator (DavidLee528/garak:proxy_dev); AutoFD: LLM as smart failure detector, a generic multi-level failure detection algorithm (Research Project).
- jdyjjj/All-in-One-Gait(Prototype system of gait recognition, includes three processes: object detection, silhouette segmentation, and gait feature extraction. It involves three deep learning models: ByteTrack, PaddleSeg, and GaitBase. My main contributions were algorithm performance optimization.)
  - **Merged**: **68% performance enhance** of silhouette segmentation module#12.
- EasyJailbreak/EasyJailbreak(Hybrid Jailbreak Attack Prompt Generation Framework)
  - **Merged**: Python interpreter version bug fix#16#17#18

Publications

- **Tianhao Li**, Weizhi Ma, Yujia Zheng, Xinchao Fan, Guangcan Yang, Ying Li, Lijun Wang, Zhengping Li*. A Survey on Gait Recognition Against Occlusion: Taxonomy, Dataset and Methodology *(Status: Undergoing major revision by PeerJ Computer Science, JCR-Q2, IF=4.1)*
- **Tianhao Li**, Yujia Zheng, Weizhi Ma, Guangshuo Wang, Zhengping Li, and Lijun Wang. "Trustworthy Metaverse: A Comprehensive Investigation into Security Risks and Privacy Issues in Artificial Intelligence-Extended Reality Systems." In SID Symposium Digest of Technical Papers, vol. 55, pp. 872-877. 2024.
- **Tianhao Li**, Yujia Zheng, Haoan Zhang, Weizhi Ma, and Ying Li. "Research on Real-time Early Warning Mechanism of Aviation Safety Based on Finite State Machine Underlying in QAR Stream Data." In Proceedings of the 2023 7th International Conference on Big Data and Internet of Things, pp. 216-222. 2023.
- Yujia Zheng, **Tianhao Li**, Weizhi Ma, Jiaxiang Zheng, Zhengping Li, and Lijun Wang. "Unveiling Privacy Challenges: Big Data-Driven Digital Twins in Smart City Applications." In SID Symposium Digest of Technical Papers, vol. 55, pp. 49-52. 2024.
- Yujia Zheng, **Tianhao Li**, Weizhi Ma, Zhengping Li, Lijun Wang, and Ying Li. "Automated Pricing and Replenishment Decision for Vegetable Products Based on Hybrid Machine Learning Models." In Electronics, Communications and Networks, pp. 81-87. IOS Press, 2024.
- Weizhi Ma, Ying Li*, **Tianhao Li**, Haowei Yang, Zhengping Li, Lijun Wang, Junyu Xuan. SFSWTS: EEG Emotion Recognition Based on Spatial-frequency Shifted Windows Time Self-attention Neural Network. *(Status: Undergoing peer-review by IEEE Transactions on Affective Computing, JCR-Q1, IF=11.9)*
- Weizhi Ma, Yujia Zheng, **Tianhao Li**, Zhengping Li, Ying Li, and Lijun Wang. "A comprehensive review of deep learning in EEG-based emotion recognition: classifications, trends, and practical implications." PeerJ Computer Science 10 (2024): e2065.
- Weizhi Ma, Zhengping Li, **Tianhao Li**, Yujia Zheng, and Lijun Wang. "Application of Virtual Reality Technology in the Diagnosis and Treatment of Psychological Disorders: An Electroencephalography (EEG)-Based Approach." In SID Symposium Digest of Technical Papers, vol. 55, pp. 882-886. 2024.

Talks

- A Survey on Gait Recognition Against Occlusion: Taxonomy, Dataset and Methodology *(English Poster Presentation)* @ 2023 5th International Conference on Machine Learning and Intelligent System (MLIS'23), Macau SAR, China. [certificate]
- Automated Pricing and Replenishment Decision For Vegetable Products Based on Hybrid Machine Learning Models *(English Oral Presentation)* @ 2023 5th International Conference on Machine Learning and Intelligent System (MLIS'23), Macau SAR, China. [certificate]
- A Comprehensive Review of Deep Learning in EEG-based Emotion Recognition: Classifications, Trends, and Practical Implications *(English Oral Presentation)* @ 2023 5th International Conference on Machine Learning and Intelligent System (MLIS'23), Macau SAR, China. [certificate]
- Dive Into QAR Stream Data - A Real-time Early Warning Mechanism *(English Oral Presentation)* @ 2023 7th International Conference on Big Data and Internet of Things (BDIoT'23), Beijing, China. [certificate][slide]

Honors and Awards

- (Feb.2024, National, *Top?%*) ? Prize in 2023 Mathematical Contest In Modeling (MCM) [paper]
- (Nov.2023, National, *Top9.8%*) Second Prize in 11th Digital Media Technology and Creativity Contest [certificate]
- (Nov.2023, National, *Top12.4%*) Third Prize in 8th National Cryptography Contest [certificate][list][news]
- (Aug.2023, National, *Top13.1%*) Third Prize in 16th CISCN Security Project Contest [cert.][report][slide][video]
- (Apr.2023, National, *Top5.7%*) First Prize in 13th MathorCup Mathematical Modeling Challenge [cert.][paper]
- (Jun.2021, National, *Top15.3%*) Third prize in 12th Lanqiao Software Development and Algorithm Contest [cert.]
- (Sep.2023, Provincial) Second Prize in Contemporary Undergraduate Mathematical Contest in Modeling [cert.]
- (Jun.2023, Regional) Third Prize in 16th CISCN AWDP (Attack with Defense Plus) Contest [certificate]
- (Apr.2023, Provincial) First Prize in 14th Lanqiao Software Development and Algorithm Contest [certificate]
- (Jun.2022, Regional) Third Prize in 15th CISCN CTF (Capture the Flag) Contest [certificate]
- (Apr.2021, Provincial) First Prize in 12th Lanqiao Software Development and Algorithm Contest [certificate]
- (2022-2023, *Top0.15%*) Received scholarships for outstanding students, academic, innovation, athletes, moral.
- (2021-2022, *Top2.91%*) Received scholarships for outstanding students, academic, innovation, athletes, moral.
- (2020-2021, *Top17.61%*) Received scholarships for academic, innovation.

Education

Duke University **August 2024 – May 2026 (Expected)**

Master of Science Student in Medical Physics

Research Focus: Trustworthy ML(Security, Privacy, Robustness, Fairness); Large Language Model; AI for Medicine

Core Courses: Academic Research, Modern Diagnostic Imaging Systems, Machine Learning, Deep Learning

North China University of Technology **September 2020 – June 2024**

Bachelor of Engineering in Information Security *GPA: 3.89/4 Grade: 90.88/100 transcript*

Core Courses: Applied Cryptography, Data Security, Image Processing, Principles of Computer Composition, Operating System, Reverse Engineering, Digital Forensic, Software Security, Network Attack and Defense

Honors: Outstanding Dissertation Award (Rank 1/101, Top 0.99%), Elite Student Scholarship (Rank 1/329, Top 0.30%)

Skills

Programming Language & OS: C/C++11, Python3, GNU/Linux(Ubuntu)

Teamwork & Documentation: Native Mandarin, Fluent Professional English (CET-4: 552, CET-6: 529, IELTS: 6.5), Git, SVN, LaTeX, Markdown

Software Framework: Machine Learning(PyTorch, Tensorflow, LangChain, TensorRT, OpenCV), ML Vulnerability Assessment(Garak, PyRIT, Counterfit), Cryptography(OpenSSL, GmSSL), Development(Django, BoostC++, MFC)

AI Red Teaming: M-LLM, Adversarial Attack, Prompt Injection, Insecure Plugins, Fine-tuning, RLHF, CoT, RAG

Miscellaneous: Driving (Since Sep 2020), Photography, Adobe PS/PR/AE/AI/AU

Extracurricular Activities

- Conference Attendance: ISC 2016, GeekPwn 2020, GeekCon 2023, CyberSecurity 2023, CCF NCCA 2024.
- Training Experience: Google Digital Talent Development Program (certificate), China Computer Federation (CCF) Artificial Intelligence Course (certificate), Ministry of Industry and Information Technology of the People's Republic of China Industry and Information Technology Talents Project (certificate), Rescue Skills (CPR and Care For Injures) Training of Red Cross Society of China Beijing Branch (certificate).
- Mountain Biking: See POV videos on bilibili channel. Top 9 racer in 2023 GDL Thaiwoo downhill race.
- Road/Trail Running and Hiking: Finisher in 2023 Chongli 168 Ultra Trail(Biggest in Asia) DTC-100KM race.
- Photography and Off-road Driving: Keen on wildlife photography and exploring the unknown.
- Voluntary service: Nearly 400 hours were recorded by Beijing Volunteer Association (As of July 2023)

李天昊

(+86) 13121515269 litianhao.life tianhao.li2@duke.edu 领英 谷歌学术 Orcid Github

工作经历

北京天融信网络安全技术有限公司 **2024年3月 – 2024年7月 (5个月)**

安全研究员 @ 创新技术研究部 中国，北京

- **职责**: **(i)** 支撑面向大模型的自动化红队预研项目，利用大模型微调、RLHF等技术，研究恶意提示词样本混淆变异与多层级智能失败检测等创新性技术，减少红队测试过程中对于安全专家的依赖，提升红队自动化水平与漏洞发现能效; **(ii)** 支撑机器学习模型安全评估项目原型系统落地，涉及到对于对抗攻击、投毒攻击、等算法的调试以及原型系统后端评估模块的开发。

绿盟科技集团股份有限公司 **2023年9月 – 2024年3月 (7个月)**

安全研究员实习 @ 创新研究院天枢实验室 中国，北京

- **职责**: **(i)** 设计绿盟大模型安全评估系统(NSFOCUS LSAS)核心算法与原型系统; **(ii)** 通过进攻性测试，即"红队"测试，提高绿盟风云卫大模型(NSFGPT)和其他私有LLM的安全和隐私性能; **(iii)** 进行文献综述，包括可信人工智能的研究论文和Gartner咨询报告，丰富威胁情报数据库; **(iv)** 撰写LLM安全技术白皮书。
- **影响**: 发布了绿盟大模型安全评估系统(LSAS)，绿盟风云卫大模型NSFGPT以及安全行业大模型SecLLM技术白皮书
- **导师**: 王星凯博士, 刘文懋博士

高精尖创新研究院 **2022年7月 – 2023年7月 (1年1个月)**

科研实习 @ 虚拟与增强现实技术创新中心 中国，北京

- **职责**: **(i)** 参与国家重点研发计划; **(ii)** 研究基于步态的情报、监视和侦察（ISR）系统; **(iii)** 研究基于脑电图（EEG）和Swin-Transformer的人类情绪分类算法; **(iv)** 研究基于QAR流数据的航空安全预警机制; **(v)** 研究元宇宙概念下AI-XR系统自身的威胁模型和缓解方法。
- **影响**: 共同撰写了7篇SCI/EI论文，赢得了4项国家级专业比赛，参加国际会议并做3次口头报告与1篇海报汇报。
- **导师**: 王立军院士, 李争平教授, 李颖副教授

辰创科技发展有限公司 **2022年1月 – 2022年3月 (3个月)**

C++软件开发实习 @ 技术开发部 中国, 成都

- **职责**: **(i)** 使用WASM、OpenCV、YOLOv5开发实时流视频解码和目标检测模块; **(ii)** 设计并实现基于SM2和SM3的安全密钥交换机制，用SM4加密RTSP视频流，同时使用SM4实现云端目标检测日志数据的加密存储，保证数据传输与存储安全。
- **影响**: 支撑了机场跑道异物(FOD)智能检测和管理系统的研发与发布。

项目实践

大语言模型安全漏洞评估系统 **2023年9月 – 2024年6月 (预计)**

算法研究员, 全栈开发者 开题答辩 | 中期检查 | 结题答辩

- **作为算法研究员**: 为不同的漏洞（如对抗攻击、越狱、DAN）设计了探针和检测器。支持了多模态大模型，可以进行跨文本、图像和音频领域的测试。 （大模型红队工具）
- **作为全栈开发者**: 开发了基于Python和Django的B/S平台，支持ModelOps、模型监控和隐私保护功能。 (Gartner AI TRiSM 概念性安全框架的实践)

基于步态的情报、监视与侦察(ISR)系统 **2023年5月 – 2023年8月**

团队领导者, 算法工程师, 后端开发者 演示视频1 | 演示视频2 | 介绍页

- **作为团队领导者**: 根据成员能力优势分配团队角色，进行系统安全能力建设与技术架构设计，编写技术报告。
- **作为算法工程师**: 实现步态识别流水线，使用PyTorch训练和评估模型。建设安全能力：身份验证、跟踪和近源渗透预警。
- **作为后端开发者**: 开发加密模块与网络模块，基于SM2/SM3/SM4算法，在公开信道实现安全密钥分发和数据加解密。

开源活动

- leondz/garak(生成式人工智能红队工具，大模型安全领域的Metasploit，本人贡献主要包括多模态以及智能化红队能力建设)
  - **已经合并**: Python解释器版本错误修复#296; Replicate大模型生成器异常修复#401; Detector检测器基类增加异常处理逻辑的增加#566; 针对LLaVA的多模态越狱攻击#587 (使其成为**世界上首个**具备多模态红队能力的工具). (v0.9.0.9, v0.9.0.11, v0.9.0.13)
  - **正在进行**: 多模态红队提示词探针架构升级 (#658); 生成器网络代理特性 (DavidLee528/garak:proxy_dev); AutoFD: 大模型即判官，一种通用的多层级失败检测算法 （研究项目）。
- jdyjjj/All-in-One-Gait(步态识别原型系统，包含目标检测、剪影分割、步态特征提取三个流程，涉及ByteTrack, PaddleSeg, GaitBase三个深度学习模型，本人贡献主要包括算法性能优化)
  - **已经合并**: 剪影分隔模块**68%性能优化**#12
- EasyJailbreak/EasyJailbreak(混合越狱攻击提示词生成框架)
  - **已经合并**: Python解释器版本错误修复#16#17#18

文章发表

- **Tianhao Li**, Weizhi Ma, Yujia Zheng, Xinchao Fan, Guangcan Yang, Ying Li, Lijun Wang, Zhengping Li*. A Survey on Gait Recognition Against Occlusion: Taxonomy, Dataset and Methodology *(Status: Undergoing major revision by PeerJ Computer Science, JCR-Q2, IF=4.1)*
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- **Tianhao Li**, Yujia Zheng, Haoan Zhang, Weizhi Ma, and Ying Li. "Research on Real-time Early Warning Mechanism of Aviation Safety Based on Finite State Machine Underlying in QAR Stream Data." In Proceedings of the 2023 7th International Conference on Big Data and Internet of Things, pp. 216-222. 2023.
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公开演讲

- A Survey on Gait Recognition Against Occlusion: Taxonomy, Dataset and Methodology *(English Poster Presentation)* @ 2023第五届机器学习与智能系统国际会议 (MLIS'23), 中国, 澳门特别行政区 [证书]
- Automated Pricing and Replenishment Decision For Vegetable Products Based on Hybrid Machine Learning Models *(English Oral Presentation)* @ 2023第五届机器学习与智能系统国际会议 (MLIS'23), 中国, 澳门特别行政区 [证书]
- A Comprehensive Review of Deep Learning in EEG-based Emotion Recognition: Classifications, Trends, and Practical Implications *(English Oral Presentation)* @ 2023第五届机器学习与智能系统国际会议 (MLIS'23), 中国, 澳门特别行政区 [证书]
- Dive Into QAR Stream Data - A Real-time Early Warning Mechanism *(English Oral Presentation)* @ 2023第七届大数据与物联网国际会议 (BDIoT'23), 中国, 北京 [证书][幻灯片]

荣誉奖项

- (2024年02月, 国家级, *Top?%*) 2024美国大学生数学建模竞赛(MCM)?奖 [论文]
- (2023年11月, 国家级, *Top9.8%*) 第十一届全国大学生数字媒体科技作品及创意竞赛二等奖 [证书]
- (2023年11月, 国家级, *Top12.4%*) 第八届全国密码技术竞赛三等奖 [证书][名单][新闻]
- (2023年08月, 国家级, *Top13.1%*) 第十六届全国大学生信息安全竞赛作品赛三等奖 [证书][报告][幻灯片][演示][新闻]
- (2023年04月, 国家级, *Top5.7%*) 第十三届MathorCup数学建模挑战赛一等奖 [证书][论文]
- (2021年06月, 国家级, *Top15.3%*) 第十二届蓝桥杯全国软件和信息技术专业人才大赛三等奖 [证书]
- (2023年09月, 省部级) 全国大学生数学建模竞赛北京赛区甲组二等奖 [证书]
- (2023年06月, 区域级) 第十六届全国大学生信息安全竞赛实践赛(AWDP)三等奖 [证书]
- (2023年04月, 省部级) 第十四届蓝桥杯全国软件和信息技术专业人才大赛一等奖 [证书]
- (2022年06月, 区域级) 第十五届全国大学生信息安全竞赛实践赛(CTF)三等奖 [证书]
- (2021年04月, 省部级) 第十二届蓝桥杯全国软件和信息技术专业人才大赛一等奖 [证书]
- (2022-2023学年, *Top0.15%*) 荣获优秀学生、学习、创新实践、体育、思想品德奖学金。
- (2021-2022学年, *Top2.91%*) 荣获优秀学生、学习、创新实践、体育、思想品德奖学金。
- (2020-2021学年, *Top17.61%*) 荣获学习、创新实践奖学金。

教育背景

杜克大学 **2024年8月 – 2026年5月 (预计)**

理学硕士在读 - 医学物理

研究方向: 可信机器学习(安全性, 隐私性, 鲁棒性, 公平性)；大语言模型；医学人工智能

核心课程: 学术研究, 现代诊断影像系统, 机器学习, 深度学习

北方工业大学 **2020年9月 – 2024年6月**

工学学士 - 信息安全 *绩点: 3.89/4 成绩: 90.88/100* 成绩单

核心课程: 应用密码学, 数据安全, 数字图像处理, 计算机组成原理, 操作系统, 逆向工程, 数字取证, 软件安全, 网络攻防

荣誉奖项: 优秀毕业论文 (排名1/101, Top 0.99%), 一等优秀学生奖学金 (排名1/329, Top 0.30%)

专业技能

编程语言与操作系统: C/C++11, Python3, GNU/Linux(Ubuntu), Microsoft Windows

团队合作与文档撰写: 普通话, 专业英语 (四级：552，六级：529，雅思：6.5), Git, SVN, LaTeX, Markdown

软件框架: 机器学习(PyTorch, Tensorflow, LangChain, TensorRT, OpenCV), 机器学习漏洞评估(Garak, PyRIT, Counterfit), 应用密码学(OpenSSL, GmSSL), 开发(Django, BoostC++, MFC)

AI红队: 多模态大模型，对抗攻击，越狱攻击，提示注入，插件利用，大模型微调，RLHF，CoT，RAG

杂项: 驾驶(自2020年9月), 摄影, Adobe PS/PR/AE/AI/AU

课外活动

- 参会经历：ISC 2016，GeekPwn 2020，GeekCon 2023，CyberSecurity 2023，CCF NCCA 2024。
- 培训经历：谷歌(Google)数字人才培养计划(证书)，中国计算机学会(CCF)群星计划(证书)，中华人民共和国工业和信息化部工信人才能力培养计划(证书)，北京市红十字会救护技能培训-心肺复苏与创伤救护(证书)。
- 山地自行车: 可以访问B站主页观看视频。2023GDL太舞速降赛排名前九。
- 跑步与徒步: 荣获2023崇礼168超级越野赛（亚洲最大）DTC百公里组完赛奖。
- 摄影与驾驶: 热衷于野生动物摄影和探索未知世界。
- 志愿服务: 志愿北京平台记录近400小时。